

# ENGINEERING ECONOMICS - COMPLETE FORMULA REFERENCE GUIDE

For Midterm Exam | Fundamentals of Engineering Economics (ECO 290)

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## HOW TO USE THIS GUIDE

Each formula includes: 1. **Formula** in boxed LaTeX format 2. **Variable definitions** with full forms 3. **When to use** (conditions/scenarios) 4. **Example statement** (typical exam question) 5. **Priority rating** ( = highest)

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## CHAPTER 1: INTRODUCTION & EFFICIENCY

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### 1. Technical Efficiency

Formula:

$$\text{Technical Efficiency} = \frac{\text{Output}}{\text{Input}}$$

Variables:

- **Output** = Physical output produced (kW, units, hp)
- **Input** = Physical input consumed (fuel, raw material)

When to Use:

| Condition                | Use This Formula                         |
|--------------------------|------------------------------------------|
| Physical system analysis | Analyzing engines, machines, furnaces    |
| Equipment performance    | Comparing different equipment efficiency |
| Energy conversion        | Power plants, motors, generators         |

Example Statement:

*“A diesel engine consumes 5 liters/hour and produces 100 kW. If 1 liter fuel = 25 kW potential, find technical efficiency.”*

Priority: (Definition, less calculation-heavy)

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### 2. Economic Efficiency (Productivity)

Formula:

$$\text{Economic Efficiency} = \frac{\text{Annual Revenue}}{\text{Annual Expenses}}$$

**Variables:**

- **Revenue** = Total annual income (Rs., \$)
- **Expenses** = Total annual costs (Rs., \$)

**When to Use:**

| Condition                | Use This Formula                             |
|--------------------------|----------------------------------------------|
| Business performance     | Overall company profitability                |
| Comparing alternatives   | Which investment is more economically viable |
| Productivity measurement | How efficiently resources are used           |

**Example Statement:**

*“A business generates Rs. 50 lakhs revenue with Rs. 40 lakhs expenses. Calculate economic efficiency.”*

**Priority: (Important concept)****3. Marginal Cost (MC)****Formula:**

$$MC = \frac{\Delta TC}{\Delta Q} = \frac{TC_2 - TC_1}{Q_2 - Q_1}$$

**Variables:**

- **MC** = Marginal Cost (Rs./unit, \$/unit) - cost of ONE MORE unit
- **$\Delta TC$**  = Change in Total Cost (TC - TC )
- **$\Delta Q$**  = Change in Quantity (Q - Q )
- **TC** = Total Cost (Rs., \$)
- **Q** = Quantity (units)

**When to Use:**

| Condition                  | Use This Formula                           |
|----------------------------|--------------------------------------------|
| “Find cost of next unit”   | Production decision making                 |
| “Should we produce more?”  | Compare MC with MR for profit maximization |
| Given TC at two quantities | Calculate per-unit cost change             |

**Example Statement:**

*“Total cost is Rs. 1000 at 10 units and Rs. 1200 at 12 units. Find marginal cost.”*

**Priority: (Core concept, frequently appears)**

#### 4. Marginal Revenue (MR)

Formula:

$$MR = \frac{\Delta TR}{\Delta Q} = \frac{TR_2 - TR_1}{Q_2 - Q_1}$$

Variables:

- **MR** = Marginal Revenue (Rs./unit) - revenue from ONE MORE unit
- **$\Delta TR$**  = Change in Total Revenue ( $TR_2 - TR_1$ )
- **$\Delta Q$**  = Change in Quantity sold
- **TR** = Total Revenue (Rs., \$)

When to Use:

| Condition                    | Use This Formula               |
|------------------------------|--------------------------------|
| “Revenue from next sale”     | Pricing decisions              |
| “Maximize profit”            | Set production where $MR = MC$ |
| Given TR at two sales levels | Calculate per-unit revenue     |

Example Statement:

*“Revenue is Rs. 5000 at 100 units and Rs. 5400 at 108 units. Find marginal revenue.”*

Priority: (Paired with MC)

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#### 5. Sunk Cost (Concept)

Rule:

|                                                   |
|---------------------------------------------------|
| Sunk Cost $\Rightarrow$ IGNORE in future analysis |
|---------------------------------------------------|

When to Use:

| Condition            | Use This Rule                              |
|----------------------|--------------------------------------------|
| Replacement analysis | Don't consider original purchase price     |
| Equipment upgrade    | Use current market value only              |
| Past investments     | Focus on incremental future costs/benefits |

Example Statement:

*“Equipment bought 3 years ago for Rs. 1,00,000 now worth Rs. 60,000. Which value for replacement analysis?”*

**Answer:** Rs. 60,000 (ignore sunk cost of Rs. 1,00,000)

Priority: (Concept, not calculation)

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## 6. Opportunity Cost (Concept)

Formula:

$$\text{Opportunity Cost of A} = \text{Benefit of Best Alternative B}$$

When to Use:

| Condition                     | Use This Concept          |
|-------------------------------|---------------------------|
| Choosing between alternatives | What are you giving up?   |
| Resource allocation           | Limited budget decisions  |
| “Should we invest in X or Y?” | Compare foregone benefits |

Example Statement:

*“Rs. 10 billion for Health or Education. Choosing Health, what’s opportunity cost?”*

**Answer:** Educational benefits foregone

Priority: (Concept)

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## CHAPTER 2: COST & BREAK-EVEN ANALYSIS

### 7. Total Sales Revenue (S or TR)

Formula:

$$S = s \times Q$$

Variables:

- **S (or TR)** = Total Sales/Total Revenue (Rs., \$)
- **s** = Selling price per unit (Rs./unit, \$/unit)
- **Q** = Quantity sold (units)

When to Use:

| Condition                | Use This Formula         |
|--------------------------|--------------------------|
| “Find total sales”       | Given price and quantity |
| Revenue calculation      | Any sales income problem |
| Building breakeven table | Calculate sales column   |

Example Statement:

*“Selling price = \$50/unit, 100 units sold. Find total sales.”*

Priority: (Basic, used everywhere)

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## 8. Total Cost (TC)

Formula:

$$TC = TVC + TFC = (v \times Q) + FC$$

Variables:

- **TC** = Total Cost (Rs., \$)
- **TVC** = Total Variable Cost (Rs., \$)
- **TFC (or FC)** = Total Fixed Cost / Fixed Cost (Rs., \$)
- **v** = Variable cost per unit (Rs./unit)
- **Q** = Quantity produced (units)

When to Use:

| Condition                    | Use This Formula    |
|------------------------------|---------------------|
| “Find total production cost” | Given FC, v, and Q  |
| Cost at any output level     | Breakeven analysis  |
| Building cost table          | Calculate TC column |

Example Statement:

“Fixed cost = \$500, variable cost = \$30/unit,  $Q = 25$ . Find total cost.”

Priority: (Essential)

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## 9. Profit

Formula:

$$\text{Profit} = S - TC = Q(s - v) - FC$$

Variables:

- **Profit** = Net Profit (positive) or Loss (negative)
- **S** = Total Sales
- **TC** = Total Cost
- **Q** = Quantity
- **s** = Selling price per unit
- **v** = Variable cost per unit
- **FC** = Fixed Cost

When to Use:

| Condition                  | Use This Formula                   |
|----------------------------|------------------------------------|
| “Find profit at Q units”   | Any profitability question         |
| “Is there profit or loss?” | Compare at different output levels |
| Verify breakeven point     | Profit should = 0 at BEQ           |

Example Statement:

“ $s = \$50$ ,  $v = \$30$ ,  $FC = \$500$ ,  $Q = 30$ . Find profit.”

Priority: (Most common question)

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## 10. Breakeven Quantity (BEQ)

Formula:

$$BEQ = \frac{FC}{s - v}$$

Variables:

- **BEQ** = Breakeven Quantity (units) - minimum units to cover all costs
- **FC** = Fixed Cost (Rs., \$)
- **s** = Selling price per unit
- **v** = Variable cost per unit
- **(s - v)** = Contribution per unit

When to Use:

| Condition                         | Use This Formula               |
|-----------------------------------|--------------------------------|
| "Find breakeven point"            | <b>#1 MOST COMMON QUESTION</b> |
| "Minimum units needed"            | Determine production targets   |
| "When do we start making profit?" | Units where profit = 0         |
| Given FC, s, v                    | Almost every cost problem      |

Example Statement:

*"FC = \$500, s = \$50/unit, v = \$30/unit. Find breakeven quantity."*

Answer Template:

$$BEQ = 500 / (50 - 30) = 500 / 20 = 25 \text{ units}$$

Priority: **HIGHEST PRIORITY**

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## 11. Breakeven Sales

Formula:

$$\text{Breakeven Sales} = BEQ \times s = \frac{FC \times s}{s - v}$$

Variables:

- **Breakeven Sales** = Sales revenue at breakeven (Rs., \$)
- **BEQ** = Breakeven Quantity
- **s** = Selling price per unit

When to Use:

| Condition                      | Use This Formula                |
|--------------------------------|---------------------------------|
| “Find breakeven sales revenue” | Want answer in Rs./\$ not units |
| “Minimum sales needed”         | Revenue-based breakeven         |

**Example Statement:**

*“BEQ = 25 units, s = \$50. Find breakeven sales.”*

**Priority:** (Alternative to BEQ)

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## 12. Contribution

**Formula:**

$$\text{Contribution per unit} = s - v$$

$$\text{Total Contribution} = Q(s - v)$$

**Variables:**

- **Contribution** = Amount per unit for FC and profit
- **s** = Selling price per unit
- **v** = Variable cost per unit
- **Q** = Quantity

**When to Use:**

| Condition                            | Use This Formula         |
|--------------------------------------|--------------------------|
| Building block for BEQ               | BEQ = FC / Contribution  |
| “How much per unit for fixed costs?” | Profitability per unit   |
| Calculating P/V ratio                | P/V = Contribution/Sales |

**Example Statement:**

*“s = \$50, v = \$30. Find contribution per unit.”*

**Priority:** (Component of other formulas)

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## 13. Margin of Safety

**Formula:**

$$\text{Margin of Safety} = \text{Actual Sales} - \text{BE Sales}$$

$$\text{MOS \%} = \frac{\text{Actual Sales} - \text{BE Sales}}{\text{Actual Sales}} \times 100$$

**Variables:**

- **MOS** = Margin of Safety - safety buffer above breakeven
- **Actual Sales** = Current sales revenue
- **BE Sales** = Breakeven sales

**When to Use:**

| Condition                        | Use This Formula           |
|----------------------------------|----------------------------|
| “How much can sales drop?”       | Risk assessment            |
| “Safety cushion above breakeven” | Investment safety analysis |
| Given actual and BE sales        | Compare alternatives       |

**Example Statement:**

*“Actual sales = \$2500, BE sales = \$1250. Find margin of safety and %.”*

**Priority:** (Risk analysis)

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**14. P/V Ratio (Profit-Volume Ratio)****Formula:**

$$\text{P/V Ratio} = \frac{s - v}{s} \times 100 = \frac{\text{Contribution}}{\text{Sales}} \times 100$$

**Variables:**

- **P/V Ratio** = Profit-Volume Ratio (%) - contribution per rupee of sales
- **s** = Selling price per unit
- **v** = Variable cost per unit

**When to Use:**

| Condition                          | Use This Formula                                               |
|------------------------------------|----------------------------------------------------------------|
| “Contribution per dollar of sales” | Profitability metric                                           |
| Alternative BEQ calculation        | $\text{BEQ} = \text{FC} / (\text{P/V Ratio} \times s)$         |
| “Find profit at sales level X”     | $\text{Profit} = (\text{Sales} \times \text{P/V}) - \text{FC}$ |

**Example Statement:**

*“s = \$50, v = \$30. Find P/V ratio.”*

**Priority:** (Very important, multiple uses)

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**Related P/V Formulas:**

**Profit using P/V Ratio:**

$$\text{Profit} = (\text{Sales} \times \text{P/V Ratio}) - FC$$

**When to use:** Calculate profit from sales amount (not quantity)

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**Required Sales for Target Profit:**

$$\text{Required Sales} = \frac{FC + \text{Desired Profit}}{\text{P/V Ratio}}$$

**When to use:** “What sales needed for Rs.  $X$  profit?”

**Example:** “ $FC = \$500$ ,  $P/V = 40\%$ , desired profit =  $\$300$ . Find required sales.”

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## CHAPTER 3: DEMAND & SUPPLY

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### 15. Equilibrium Condition

**Formula:**

$$Q_d = Q_s$$

**At equilibrium:**

$$P_e, Q_e \text{ where demand} = \text{supply}$$

**Variables:**

- **Qd** = Quantity Demanded
- **Qs** = Quantity Supplied
- **Pe** = Equilibrium Price
- **Qe** = Equilibrium Quantity

**When to Use:**

| Condition                         | Use This Formula              |
|-----------------------------------|-------------------------------|
| “Find equilibrium price/quantity” | Market clearing conditions    |
| Given demand and supply equations | Set $Q_d = Q_s$ , solve for P |
| “Find market equilibrium”         | Where no shortage/surplus     |

**Example Statement:**

*“Demand:  $Q_d = 100 - 2P$ . Supply:  $Q_s = 20 + 3P$ . Find equilibrium.”*

**Solution Template:**

$$100 - 2P = 20 + 3P$$

$$80 = 5P$$

$$P_e = 16$$

$$Q_e = 100 - 2(16) = 68$$

**Priority:** (Core market analysis)

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## 16. Market Disequilibria

**Excess Demand (Shortage):**

$$Q_d > Q_s \Rightarrow \text{Price } \uparrow$$

**Excess Supply (Surplus):**

$$Q_s > Q_d \Rightarrow \text{Price } \downarrow$$

**When to Use:**

| Condition                                  | Use This Concept             |
|--------------------------------------------|------------------------------|
| “Will price rise or fall?”                 | Predict price movement       |
| Market not in equilibrium                  | Shortage/surplus situations  |
| “ $Q_d = 100$ , $Q_s = 60$ at $P = \$10$ ” | Calculate excess demand = 40 |

**Priority:** (Conceptual)

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## CHAPTER 4: ELASTICITY

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### 17. Price Elasticity of Demand (PED) - Basic

**Formula:**

$$PED = \left| \frac{\% \Delta Q_d}{\% \Delta P} \right| = \left| \frac{\Delta Q_d / Q_d}{\Delta P / P} \right|$$

**Variables:**

- **PED** = Price Elasticity of Demand (unitless, always positive)
- **$\Delta Q_d$**  = Change in Quantity Demanded ( $Q_d - Q_d$ )
- **$Q_d$**  = Original Quantity Demanded
- **$\Delta P$**  = Change in Price ( $P - P$ )
- **P** = Original Price

**When to Use:**

| Condition               | Use This Formula            |
|-------------------------|-----------------------------|
| “Find price elasticity” | Basic elasticity question   |
| Given % changes         | “P rises 10%, Qd falls 15%” |
| Simple calculations     | Quick estimates             |

**Example Statement:**

*“Price rises by 10%, quantity demanded falls by 15%. Calculate PED.”*

**Priority: HIGHEST**

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## 18. PED - Midpoint Method

**Formula:**

$$PED = \frac{|Q_2 - Q_1|}{|P_2 - P_1|} \times \frac{(P_2 + P_1)/2}{(Q_2 + Q_1)/2}$$

**Variables:**

- **Q** , **Q** = Quantity at two points
- **P** , **P** = Price at two points
- **(Q + Q)/2** = Midpoint of quantity (average)
- **(P + P)/2** = Midpoint of price (average)

**When to Use:**

| Condition                              | Use This Formula                          |
|----------------------------------------|-------------------------------------------|
| “ <b>Given two (P,Q) points</b> ”      | <b>#1 MOST COMMON ELASTICITY QUESTION</b> |
| “Calculate elasticity between A and B” | Direction-independent answer              |
| Exact values given                     | More accurate than basic method           |

**Example Statement:**

*“At P = \$200, Q = 12. At P = \$250, Q = 8. Find PED.”*

**Solution Template:**

$$P_{mid} = (200+250)/2 = 225$$

$$Q_{mid} = (12+8)/2 = 10$$

$$\% \Delta P = |250-200|/225 = 50/225 = 22.2\%$$

$$\% \Delta Q = |8-12|/10 = 4/10 = 40\%$$

$$PED = 40/22.2 = 1.8$$

**Priority: HIGHEST PRIORITY**

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## 19. Total Revenue (TR)

Formula:

$$TR = P \times Q$$

Variables:

- **TR** = Total Revenue (Rs., \$)
- **P** = Price per unit
- **Q** = Quantity sold

When to Use:

| Condition                 | Use This Formula                            |
|---------------------------|---------------------------------------------|
| Analyzing revenue changes | Compare TR before/after price change        |
| With elasticity           | Determine if price change increases revenue |

Priority: (Used with PED)

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## 20. TR-Elasticity Relationships

Formula:

$$\begin{cases} \text{PED} < 1 \text{ (Inelastic):} & P \uparrow \Rightarrow TR \uparrow \\ \text{PED} > 1 \text{ (Elastic):} & P \uparrow \Rightarrow TR \downarrow \\ \text{PED} = 1: & P \text{ change} \Rightarrow TR \text{ unchanged} \end{cases}$$

When to Use:

| Condition                           | Use This Rule                                                          |
|-------------------------------------|------------------------------------------------------------------------|
| “Should we raise price?”            | <b>CRITICAL BUSINESS DECISION</b>                                      |
| “Will price increase help revenue?” | Check elasticity first                                                 |
| Pricing strategy                    | Inelastic $\rightarrow$ raise price, Elastic $\rightarrow$ lower price |

Example Statement:

“Demand is inelastic ( $PED = 0.8$ ). Price increases from \$1 to \$3. Quantity falls 100 $\rightarrow$ 80. Verify TR increases.”

Priority: **HIGHEST** (Strategy)

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## 21. Income Elasticity (YED)

Formula:

$$YED = \frac{\% \Delta Q_d}{\% \Delta Y} = \frac{\Delta Q_d / Q_d}{\Delta Y / Y}$$

**Variables:**

- **YED** = Income Elasticity of Demand
- **$\Delta Q_d$**  = Change in Quantity Demanded
- **$Q_d$**  = Original Quantity
- **$\Delta Y$**  = Change in Income
- **Y** = Original Income

**When to Use:**

| Condition                           | Use This Formula                      |
|-------------------------------------|---------------------------------------|
| “Effect of income change on demand” | Income changes, find demand impact    |
| “Classify good as luxury/necessity” | $YED > 1$ = luxury, $0-1$ = necessity |
| “Is this inferior good?”            | $YED < 0$ = inferior                  |

**Example Statement:**

“Income rises from Rs. 500 to Rs. 600. Demand increases from 20 to 25 units. Find YED and classify good.”

**Solution Template:**

$\Delta Y = 100$ ,  $Y = 500$

$\Delta Q_d = 5$ ,  $Q_d = 20$

$YED = (5/20) / (100/500) = 0.25 / 0.20 = 1.25$

$YED > 1 \rightarrow$  Luxury good

**Priority:** (Common)

**22. Cross-Price Elasticity (XED)****Formula:**

$$XED = \frac{\% \Delta Q_{d,A}}{\% \Delta P_B} = \frac{\Delta Q_{d,A} / Q_{d,A}}{\Delta P_B / P_B}$$

**Variables:**

- **XED** = Cross-Price Elasticity (keep sign!)
- **$\Delta Q_{d,A}$**  = Change in demand for good A
- **$Q_{d,A}$**  = Original demand for good A
- **$\Delta P_B$**  = Change in price of good B
- **$P_B$**  = Original price of good B

**When to Use:**

| Condition                                 | Use This Formula                     |
|-------------------------------------------|--------------------------------------|
| “Are A and B substitutes or complements?” | <b>RELATIONSHIP QUESTION</b>         |
| Two related goods                         | Price of one affects demand of other |
| Competitive analysis                      | Market relationships                 |

**Interpretation:**

- **XED > 0:** Substitutes (Tea & Coffee)
- **XED < 0:** Complements (Cars & Gas)
- **XED = 0:** Unrelated

**Example Statement:**

*“Rice price rises Rs. 2000→2500. Wheat demand rises 40→50 kg. Find XED and relationship.”*

**Priority:** (Frequent)

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**23. Price Elasticity of Supply (PES)****Formula:**

$$PES = \frac{\% \Delta Q_s}{\% \Delta P} = \frac{\Delta Q_s / Q_s}{\Delta P / P}$$

**Variables:**

- **PES** = Price Elasticity of Supply
- **ΔQs** = Change in Quantity Supplied
- **Qs** = Original Quantity Supplied
- **ΔP** = Change in Price
- **P** = Original Price

**When to Use:**

| Condition                            | Use This Formula            |
|--------------------------------------|-----------------------------|
| “How responsive is supply to price?” | Supply flexibility          |
| Short-run vs long-run comparison     | Time period analysis        |
| Given price and supply data          | Calculate supply elasticity |

**Example Statement:**

*“Price rises from Rs. 10 to Rs. 12. Supply increases from 20 to 24 units. Find PES.”*

**Priority:** (Less common than demand elasticity)

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**QUICK DECISION FLOWCHART**

**Question Type → Formula to Use**

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| If Question Says...           | Use This Formula                           | Variable Priority       |
|-------------------------------|--------------------------------------------|-------------------------|
| "Find breakeven"              | <b>BEQ</b> = $FC/(s-v)$                    | FC, s, v                |
| "Two (P,Q) points given"      | <b>PED Midpoint Method</b>                 | P, P, Q, Q              |
| "Profit at Q units"           | <b>Profit</b> = $Q(s-v) - FC$              | Q, s, v, FC             |
| "Should we raise price?"      | Calculate <b>PED</b> , then <b>TR rule</b> | P, Q data               |
| "Substitutes or complements?" | <b>XED</b> (check sign)                    | QA, PB                  |
| "Luxury or necessity?"        | <b>YED</b> (check if >1 or <1)             | Q, Income               |
| "Cost of next unit"           | <b>MC</b> = $\Delta TC/\Delta Q$           | TC at two Q levels      |
| "Equilibrium price"           | Set <b>Qd</b> = <b>Qs</b>                  | Demand/supply equations |
| "Margin of safety"            | <b>Actual Sales</b> - <b>BE Sales</b>      | Sales, BEQ              |
| "P/V Ratio"                   | $(s-v)/s \times 100$                       | s, v                    |

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## PRIORITY SUMMARY

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### HIGHEST PRIORITY (Master These First)

1. **BEQ** =  $FC/(s-v)$  - Appears in 90% of cost problems
  2. **PED Midpoint Method** - Most common elasticity question
  3. **Profit** =  $Q(s-v) - FC$  - Every profitability question
  4. **TR** =  $P \times Q$  with **TR-Elasticity Rules** - Pricing strategy
  5. **Equilibrium: Qd = Qs** - Market analysis
  6. **YED** - Income effects
  7. **XED** - Substitutes/complements
  8. **P/V Ratio** =  $(s-v)/s \times 100$  - Alternative calculations
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### MEDIUM PRIORITY (Know Well)

9. **TC** =  $(v \times Q) + FC$  - Building block
  10. **MC** =  $\Delta TC/\Delta Q$  - Marginal analysis
  11. **MR** =  $\Delta TR/\Delta Q$  - Revenue analysis
  12. **Margin of Safety** - Risk assessment
  13. **PES** - Supply responsiveness
  14. **Economic Efficiency** - Productivity
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### LOWER PRIORITY (Understand Concepts)

15. **Technical Efficiency** - Definition
  16. **Sunk Cost** - Ignore in analysis
  17. **Opportunity Cost** - Conceptual
  18. **Market disequilibria** - Price movements
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## VARIABLE MASTER LIST

### Complete Alphabetical Reference

| Symbol             | Full Form                     | Typical Unit      | Used In                 |
|--------------------|-------------------------------|-------------------|-------------------------|
| <b>BEQ</b>         | Breakeven Quantity            | units             | Breakeven analysis      |
| <b>FC (or TFC)</b> | Fixed Cost / Total Fixed Cost | Rs., \$           | All cost formulas       |
| <b>MC</b>          | Marginal Cost                 | Rs./unit          | Marginal analysis       |
| <b>MR</b>          | Marginal Revenue              | Rs./unit          | Revenue analysis        |
| <b>P</b>           | Price                         | Rs./unit, \$/unit | Elasticity, equilibrium |
| <b>Pe</b>          | Equilibrium Price             | Rs./unit          | Market equilibrium      |
| <b>PED</b>         | Price Elasticity of Demand    | unitless          | Demand responsiveness   |
| <b>PES</b>         | Price Elasticity of Supply    | unitless          | Supply responsiveness   |
| <b>P/V Ratio</b>   | Profit-Volume Ratio           | %                 | Contribution metric     |
| <b>Q</b>           | Quantity                      | units             | All formulas            |
| <b>Qd</b>          | Quantity Demanded             | units             | Demand analysis         |
| <b>Qe</b>          | Equilibrium Quantity          | units             | Market equilibrium      |
| <b>Qs</b>          | Quantity Supplied             | units             | Supply analysis         |
| <b>s</b>           | Selling price per unit        | Rs./unit          | Revenue, breakeven      |
| <b>S (or TR)</b>   | Sales / Total Revenue         | Rs., \$           | Revenue calculations    |
| <b>TC</b>          | Total Cost                    | Rs., \$           | Cost analysis           |
| <b>TVC</b>         | Total Variable Cost           | Rs., \$           | Cost breakup            |
| <b>v</b>           | Variable cost per unit        | Rs./unit          | Cost, breakeven         |
| <b>XED</b>         | Cross-Price Elasticity        | unitless          | Related goods           |
| <b>Y</b>           | Income                        | Rs., \$           | Income elasticity       |
| <b>YED</b>         | Income Elasticity of Demand   | unitless          | Income effects          |
| <b>Δ</b>           | Delta (Change)                | -                 | All change calculations |

## EXAM SUCCESS STRATEGY

### Before Exam:

1. Memorize TOP 8 formulas ( rated)
2. Practice midpoint PED method 5+ times
3. Know TR-elasticity rules by heart
4. Understand when to use each formula (conditions)



### **During Exam:**

1. Read question → Identify formula needed
2. Write formula first (even if unsure)
3. Define variables in your answer
4. Show all calculation steps
5. Box/underline final answer with units

### **Common Mistakes to Avoid:**

Using wrong base for % (use midpoint!)  
Forgetting to take absolute value for PED  
Mixing up XED sign (keep it for interpretation!)  
Using  $FC/v$  instead of  $FC/(s-v)$  for BEQ  
Not showing units in final answer

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**You've got this! Good luck on your midterm!**

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